

Bill of Materials — VBF-T4R1-BOM-01

Component	Basis / formula	g/cell	kg/kWh	Source
Positive active material (NMC811)	96 wt% of pos coating (literature default)	16.168	0.901	cell/r6_t2_energy.json:layer_kg_m2 x area x 0.96
Positive conductive additive (carbon)	2 wt% (literature default)	0.337	0.019	same x 0.02
Positive binder (PVDF)	2 wt% (literature default)	0.337	0.019	same x 0.02
Negative active material (graphite)	95.5 wt% of neg coating (literature default)	10.385	0.579	cell/r6_t2_energy.json:layer_kg_m2 x area x 0.955
Negative conductive additive (carbon)	1 wt% (literature default)	0.109	0.006	same x 0.01
Negative binder (SBR/CMC)	3.5 wt% (literature default)	0.381	0.021	same x 0.035
Separator (polyolefin)	layer mass formula	0.216	0.012	cell/r6_t2_energy.json:layer_kg_m2
Electrolyte	pore volume x 1.2 g/cm3 (literature default)	6.325	0.352	pore volume = (L_pos*eps_pos + L_sep*eps_sep + L_neg*eps_neg) x area = 5.271 mL
Positive current collector (Al)	10 um x 2700 kg/m3	2.773	0.154	cell/r6_t2_energy.json:layer_kg_m2
Negative current collector (Cu)	8 um x 8960 kg/m3	7.362	0.41	cell/r6_t2_energy.json:layer_kg_m2
Enclosure / tabs	Not modeled	Not provided	Not provided	beyond pure simulation boundary
TOTAL (electrolyte excluded)	contract cell mass	38.067	2.121	cell/r6_t2_energy.json:mass_kg
TOTAL (incl. electrolyte fill)	sum of above	44.392	2.473	sum
Cell energy	simulated 1C discharge, 25 C	Not provided	Not provided	cell/r6_t2_energy.json:energy_wh = 17.9505 Wh